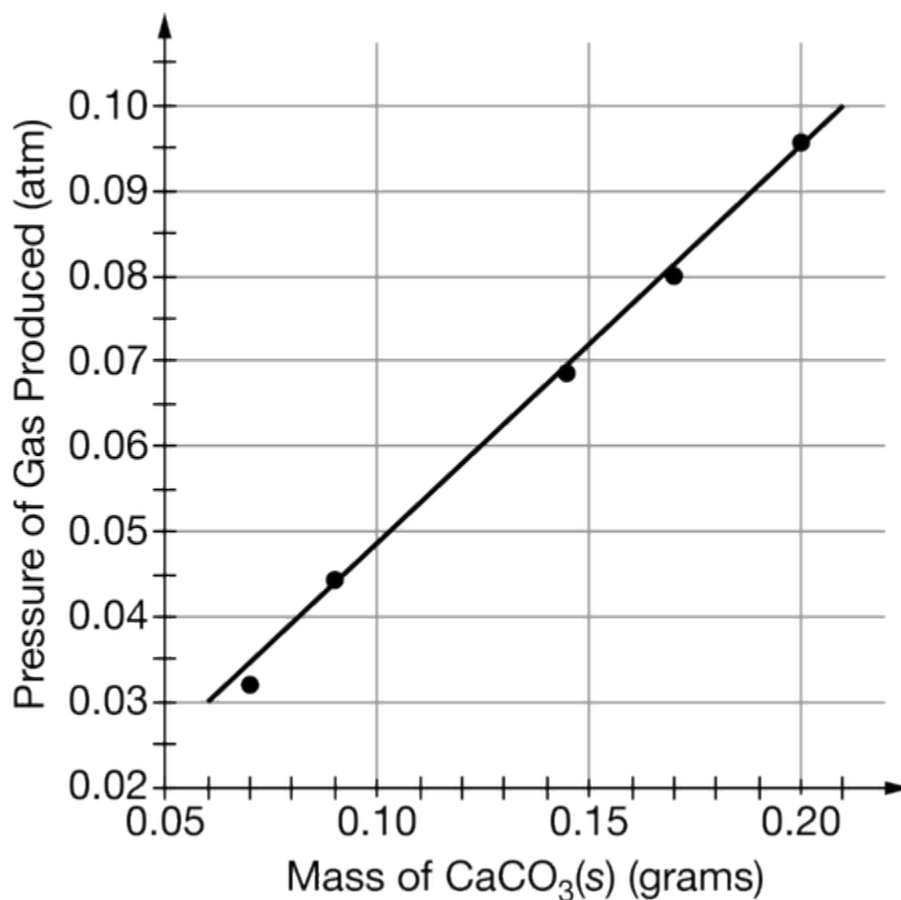


Chapter 04 Part 1: Changes and Representations

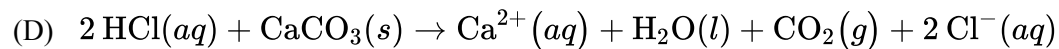
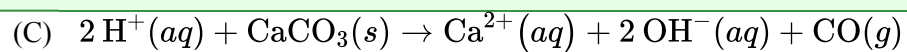
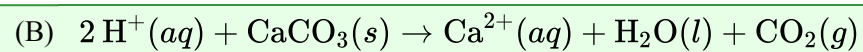
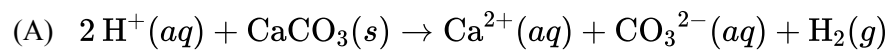
An experiment is performed to measure the mass percent of $\text{CaCO}_3(s)$ in eggshells. Five different samples of $\text{CaCO}_3(s)$ of known mass react with an excess of $2.0\text{ M HCl}(aq)$ in identical sealed, rigid reaction vessels. The pressure of the gas produced is measured with a pressure gauge attached to the reaction vessel. Since the reaction is exothermic, the reaction system is cooled to its original temperature before the pressure is recorded. The experimental data are used to create the calibration line below.



The experiment is repeated with an eggshell sample, and the experimental data are recorded in the table below.

Mass of eggshell sample	0.200 g
Pressure prior to reaction	0.800 atm
Pressure at completion of reaction	0.870 atm

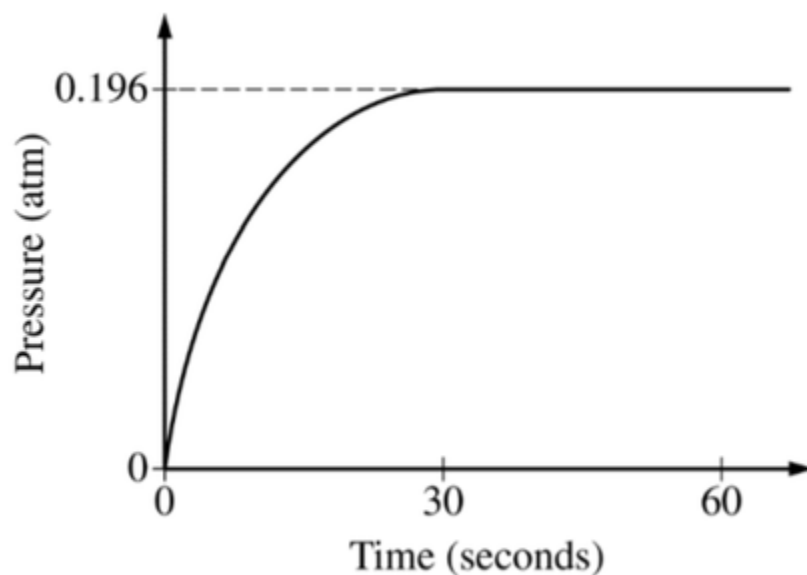
1. Which of the following equations best represents the species that react and the species that are produced when $\text{CaCO}_3(s)$ and $\text{HCl}(aq)$ are combined?

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Substance	Lewis Diagram	Boiling Point
CH_3OH	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\ddot{\text{O}}-\text{H} \\ \\ \text{H} \end{array}$	338 K
$\text{C}_2\text{H}_5\text{OH}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	351 K

Equimolar samples of $\text{CH}_3\text{OH}(l)$ and $\text{C}_2\text{H}_5\text{OH}(l)$ are placed in separate, previously evacuated, rigid 2.0 L vessels. Each vessel is attached to a pressure gauge, and the temperatures are kept at 300 K. In both vessels, liquid is observed to remain present at the bottom of the container at all times. The change in pressure inside the vessel containing $\text{CH}_3\text{OH}(l)$ is shown below.



2. Which of the following best describes the change that takes place immediately after the $\text{CH}_3\text{OH}(l)$ is introduced into the previously evacuated vessel?

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- (A) A chemical change takes place because covalent bonds are broken.
- (B) A chemical change takes place because intermolecular attractions are overcome.
- (C) A physical change takes place because covalent bonds are broken.
- (D) A physical change takes place because intermolecular attractions are overcome. ✓

3. Which of the following describes the changes in forces of attraction that occur as H_2O changes phase from a liquid to a vapor?

- (A) $\text{H}-\text{O}$ bonds break as $\text{H}-\text{H}$ and $\text{O}-\text{O}$ bonds form.
- (B) Hydrogen bonds between H_2O molecules are broken. ✓
- (C) Covalent bonds between H_2O molecules are broken.
- (D) Ionic bonds between H^+ ions and OH^- ions are broken.
- (E) Covalent bonds between H^+ ions and H_2O molecules become more effective.

4.

Mass of empty crucible	12.0 g
Mass of crucible and sample before heating	14.4 g
Mass of crucible and sample after heating	16.0 g

A student places a sample of a pure metal in a crucible and heats it strongly in air. Data from the experiment are given in the table above. The final mass was determined after the sample was cooled to room temperature. Which of the following statements related to the experiment is correct?

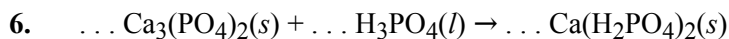
- (A) The mass of the sample decreased, so physical changes occurred as the metal first melted and then boiled out of the crucible.
- (B) The mass of the sample increased, so a chemical change occurred when bonds formed between the metal and another substance. ✓
- (C) There was nothing for the metal to react with, so only a physical change could have occurred.
- (D) The sample was only heated, so neither a physical nor a chemical change occurred.

5. $\dots \text{C}_{10}\text{H}_{12}\text{O}_4\text{S}(s) + \dots \text{O}_2(g) \rightarrow \dots \text{CO}_2(g) + \dots \text{SO}_2(g) + \dots \text{H}_2\text{O}(g)$

When the equation above is balanced and all coefficients are reduced to their lowest whole-number terms, the coefficient for $\text{O}_2(g)$ is

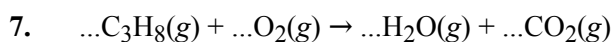
- (A) 6
- (B) 7
- (C) 12 ✓
- (D) 14
- (E) 28

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When the equation above is balanced and all coefficients are reduced to lowest whole-number terms, what is the coefficient for $\text{H}_3\text{PO}_4(l)$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5



When the equation for the reaction represented above is balanced and all coefficients are reduced to the lowest whole-number terms, the coefficient for $\text{O}_2(g)$ is

- (A) 1
- (B) 2
- (C) 3
- (D) 5
- (E) 6



8. A student mixes 20.0 g of white KCl crystals with distilled water in a beaker. After the mixture was stirred, no crystals are visible and the solution is clear. After several days, all of the water evaporates and white crystals are found in the beaker. Which of following pieces of experimental evidence would best help the student to confirm that a new compound had not been made and that only a physical change occurred?

- (A) The solution does not change color after stirring.
- (B) The KCl crystals are no longer visible after mixing with water.
- (C) There is a temperature change in the solution during the dissolving process.

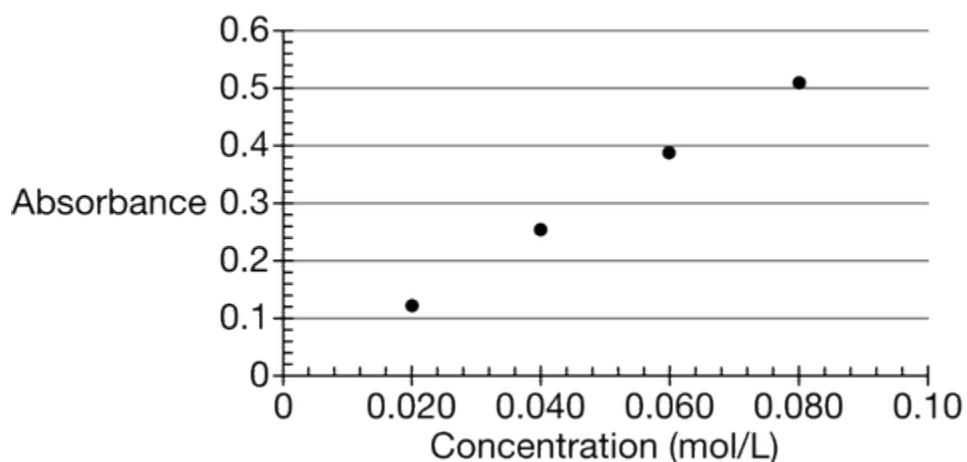
(D) After the water has evaporated, the white crystals in the beaker have a mass of 20.0 g.



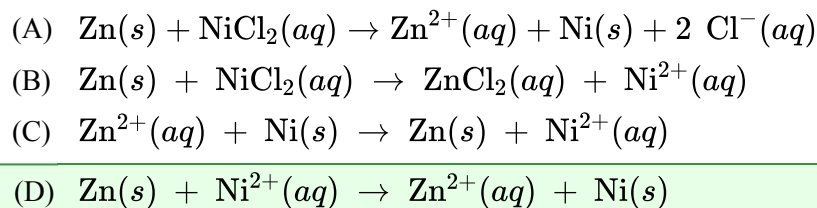
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An experiment was performed to investigate the reaction between Zn metal and $\text{Ni}^{2+}(aq)$ at different concentrations. Because the $\text{Ni}^{2+}(aq)$ ion is green, the extent of the reaction was determined using spectrophotometric analysis. Four 20.0 mL standard solutions of $\text{Ni}^{2+}(aq)$ were prepared by dissolving $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ (molar mass 240 g/mol) in water. The absorbance of each solution was measured; the results are shown both in the table below and in the following plot of the absorbance data.

Solution	$[\text{Ni}^{2+}]$	Absorbance
1	0.020	0.12
2	0.040	0.25
3	0.060	0.39
4	0.080	0.51



9. Which of the following equations best represents the reacting species in the reaction between Zn metal and $\text{NiCl}_2(aq)$?



10. $\dots \text{LiHCO}_3(aq) + \dots \text{H}_2\text{SO}_4(aq) \rightarrow \dots \text{Li}_2\text{SO}_4(aq) + \dots \text{H}_2\text{O}(l) + \dots \text{CO}_2(g)$

When the equation above is balanced and the coefficients are reduced to lowest whole-number terms, what is the coefficient of $\text{H}_2\text{O}(l)$?

Chapter 04 Part 1: Changes and Representations

(A) 1

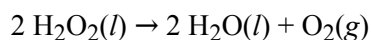
(B) 2

(C) 3

(D) 4

(E) 5

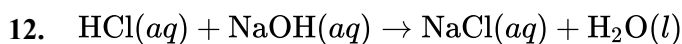
11.



The exothermic process represented above is best classified as a

- (A) physical change because a new phase appears in the products
- (B) physical change because $\text{O}_2(g)$ that was dissolved comes out of solution
- (C) chemical change because entropy increases as the process proceeds

(D) chemical change because covalent bonds are broken and new covalent bonds are formed



A student had two dilute, colorless solutions, $\text{HCl}(aq)$ and $\text{NaOH}(aq)$, which were at the same temperature. The student combined the solutions, and the reaction represented above occurred. Which of the following results would be evidence that a chemical reaction took place?

(A) The resulting solution is colorless.

(B) The temperature of the reaction mixture increases.

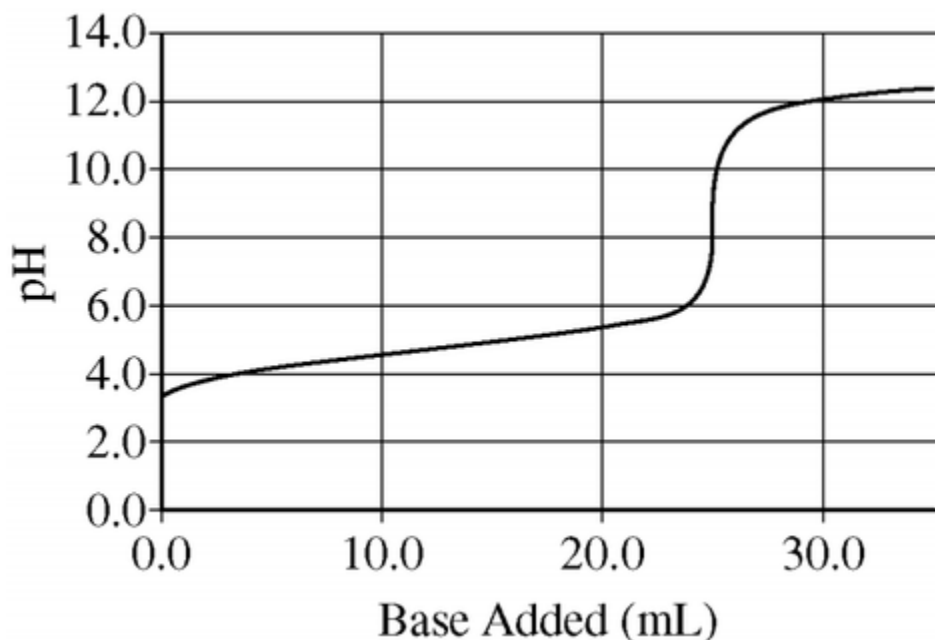
(C) The total volume of the mixture is approximately equal to the sum of the initial volumes.

(D) The resulting solution conducts electricity.

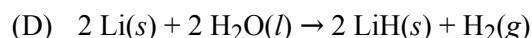
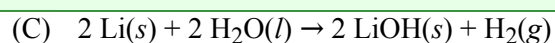
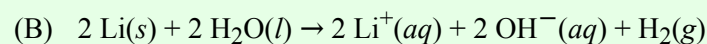
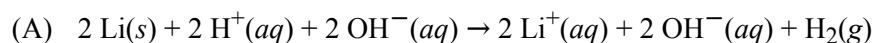
Chapter 04 Part 1: Changes and Representations

A 0.35 g sample of $\text{Li}(s)$ is placed in an Erlenmeyer flask containing 100 mL of water at 25°C . A balloon is placed over the mouth of the flask to collect the hydrogen gas that is generated.

After all of the $\text{Li}(s)$ has reacted with $\text{H}_2\text{O}(l)$, the solution in the flask is added to a clean, dry buret and used to titrate an aqueous solution of a monoprotic acid. The pH curve for this titration is shown in the diagram below



13. Which of the following is the balanced net-ionic equation for the reaction between $\text{Li}(s)$ and water?



14. A student was asked to formulate a hypothesis about what would happen if 100. mL of 0.1 M $\text{NaOH}(aq)$ at 25°C was combined with 100. mL of 0.1 M $\text{MgCl}_2(aq)$ at 25°C . Which of the following hypotheses indicates that the student thought a chemical change would occur?

(A) The volume of the resulting solution will be equal to the sum of the volumes of the original solutions.

(B) The mass of the resulting solution will be equal to the sum of the masses of the original solutions.

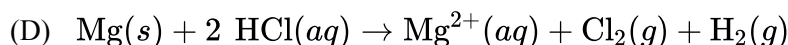
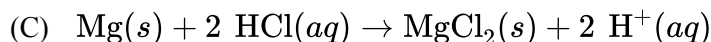
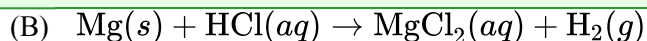
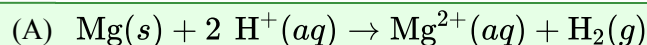
(C) The resulting solution would contain a precipitate. ✓

(D) The resulting solution will be clear.

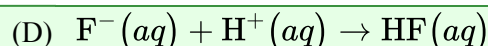
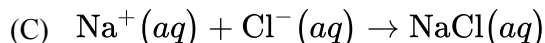
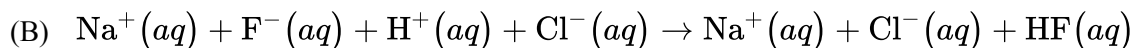
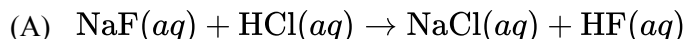
Chapter 04 Part 1: Changes and Representations

A 0.50 g sample of $\text{Mg}(s)$ was placed in a solution of $\text{HCl}(aq)$, where it reacted completely.

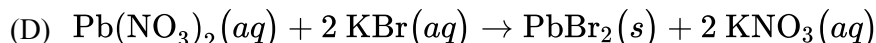
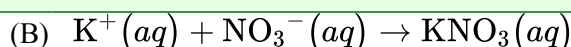
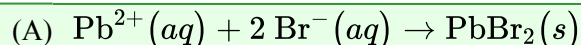
15. Which of the following equations best represents the reacting species in the reaction described above?



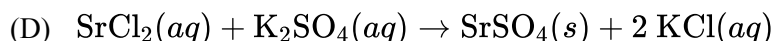
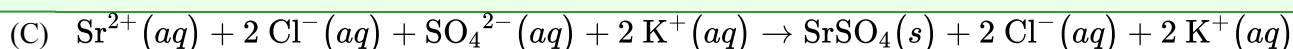
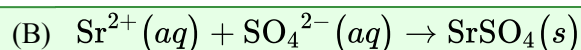
16. Which of the following is the net ionic equation for the reaction between aqueous sodium fluoride and hydrochloric acid?



17. Equal volumes of 0.2 M solutions of lead(II) nitrate and potassium bromide are combined to form lead(II) bromide as a yellow precipitate. Which of the following is the correct net ionic equation for the reaction?



18. The reaction between aqueous strontium chloride (SrCl_2) and aqueous potassium sulfate (K_2SO_4) forms a precipitate of strontium sulfate (SrSO_4). Which of the following represents the net ionic equation for the reaction?

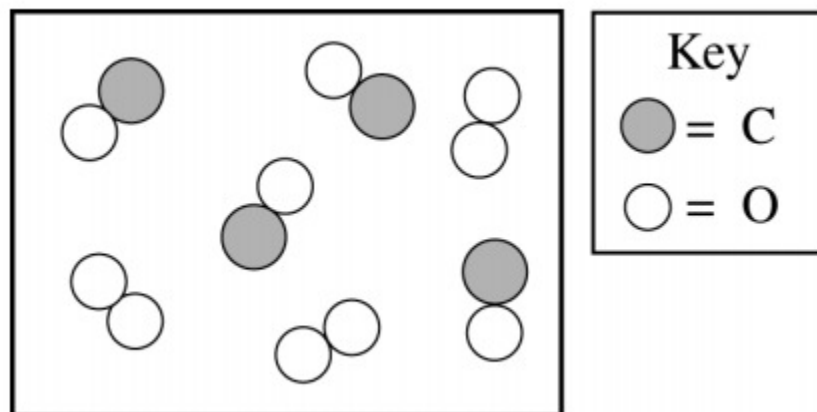


19. A 20. mL sample of 0.50 M $\text{HC}_2\text{H}_3\text{O}_2(aq)$ is titrated with 0.50 M $\text{NaOH}(aq)$. Which of the following best represents the species that react and the species produced in the reaction?

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- (A) $\text{H}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l)$
- (B) $\text{H}^+(aq) + \text{C}_2\text{H}_3\text{O}_2^-(aq) + \text{Na}^+(aq) + \text{OH}^-(aq) \rightarrow \text{H}_2\text{O}(l) + \text{NaC}_2\text{H}_3\text{O}_2(aq)$
- (C) $\text{HC}_2\text{H}_3\text{O}_2(aq) + \text{OH}^-(aq) \rightarrow \text{C}_2\text{H}_3\text{O}_2^-(aq) + \text{H}_2\text{O}(l)$ ✓
- (D) $\text{HC}_2\text{H}_3\text{O}_2(aq) + \text{NaOH}(aq) \rightarrow \text{H}_2\text{O}(l) + \text{Na}^+(aq) + \text{C}_2\text{H}_3\text{O}_2^-(aq)$

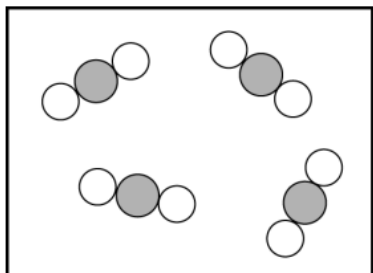
20.



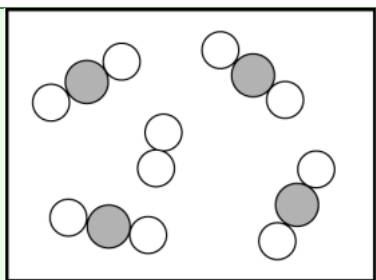
A mixture of $\text{CO}(g)$ and $\text{O}_2(g)$ is placed in a container, as shown above. A reaction occurs, forming $\text{CO}_2(g)$. Which of the following best represents the contents of the box after the reaction has proceeded as completely as possible?

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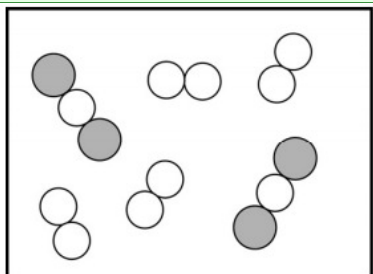
(A)



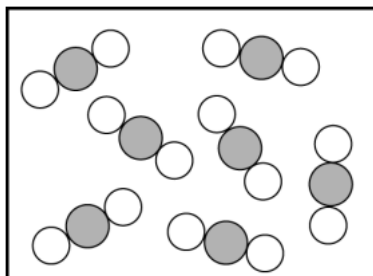
(B)



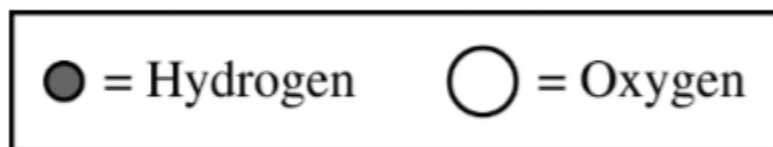
(C)



(D)

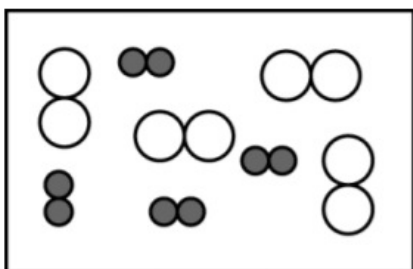


21. Which of the following particle diagrams best represents the products when four molecules of $\text{H}_2\text{O}_2(l)$ decompose into water and oxygen gas at room temperature?

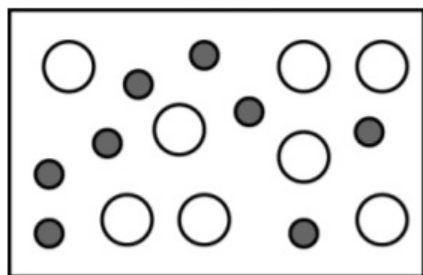


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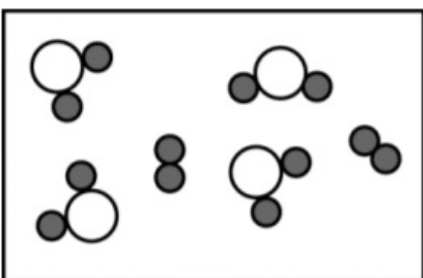
(A)



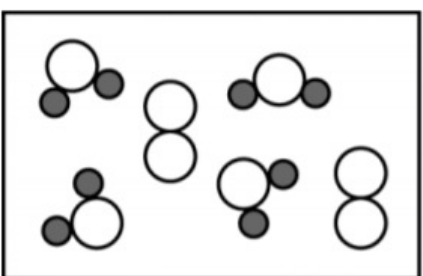
(B)



(C)

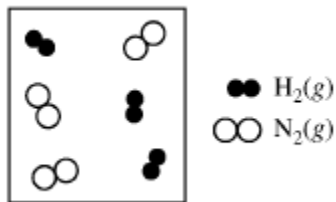


(D)

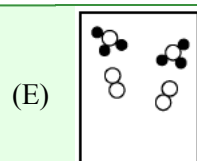
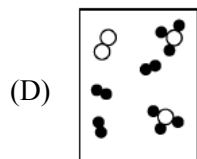
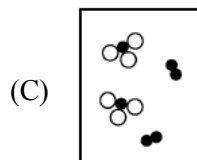
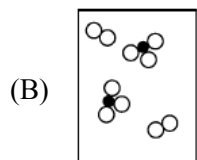
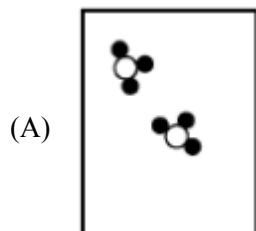
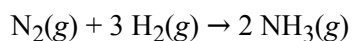


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22.

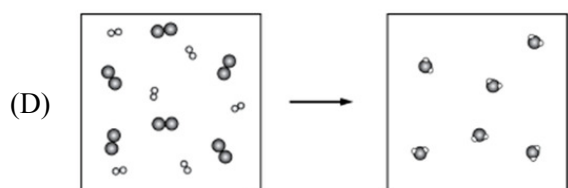
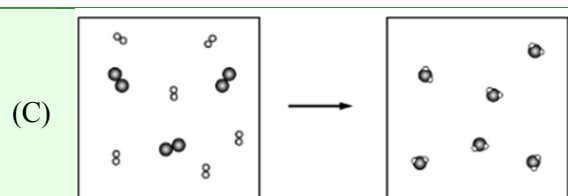
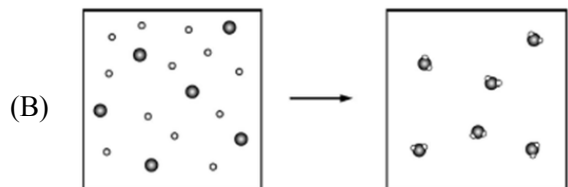
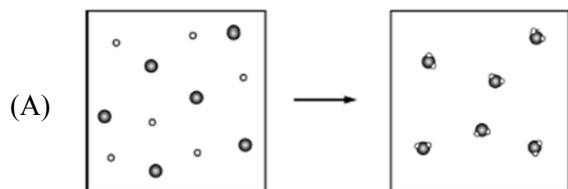


The diagram above represents $\text{H}_2(\text{g})$ and $\text{N}_2(\text{g})$ in a closed container. Which of the following diagrams would represent the results if the reaction shown below were to proceed as far as possible?



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23. Which of the following particulate diagrams best shows the formation of water vapor from hydrogen gas and oxygen gas in a rigid container at 125°C ?



24. A student was studying physical and chemical changes. The student carried out some procedures in the laboratory and recorded observations. For one of the procedures, the student concluded that a physical change took place, but not a chemical change. Which of the following could have been the results of the procedure?

(A) A cube of metal was changed into a flat sheet of metal.

(B) When two liquids at room temperature were combined in a beaker, the beaker became hot.

(C) When two clear liquids were combined, the resulting mixture was cloudy.

(D) When a colorless liquid was added to a blue liquid, the resulting solution was yellow.

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25.

Substance	Appearance	Boiling Point	Density
Water	Clear, colorless liquid	100.0 °C	1.00 g/mL
X	White, crystalline solid	1935.5 °C	2.15 g/mL
Mixture of water + X	Clear, colorless liquid	101.6 °C	1.08 g/mL

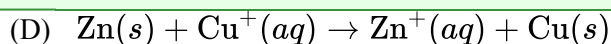
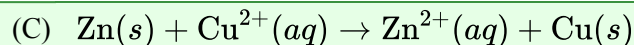
The table above summarizes data given to a student to evaluate the type of change that took place when substance X was mixed with water. The student claimed that the data did not provide enough evidence to determine whether a chemical or physical change took place and that additional tests were needed. Which of the following identifies the best way to gather evidence to support the type of change that occurred when water and X were mixed?

- (A) Measuring the melting point of the mixture of water and X
- (B) Adding another substance to the mixture of water and X to see whether a solid forms
- (C) Measuring and comparing the masses of the water, X, and the mixture of water and X
- (D) Measuring the electrical conductivities of X and the mixture of water and X



26. A student placed a piece of $\text{Zn}(s)$ in a beaker containing $\text{CuSO}_4(aq)$. After 45 minutes, the student observed a brown solid at the bottom of the beaker and that the solution had turned from blue to colorless. Which of the following balanced equations best represents only the species that react and the species that are produced in the reaction?

- (A) $2 \text{Zn}(s) + \text{CuSO}_4(aq) \rightarrow \text{Zn}_2\text{SO}_4(aq) + \text{Cu}(s)$
- (B) $\text{Zn}(s) + \text{CuSO}_4(aq) \rightarrow \text{ZnSO}_4(aq) + \text{Cu}(s)$

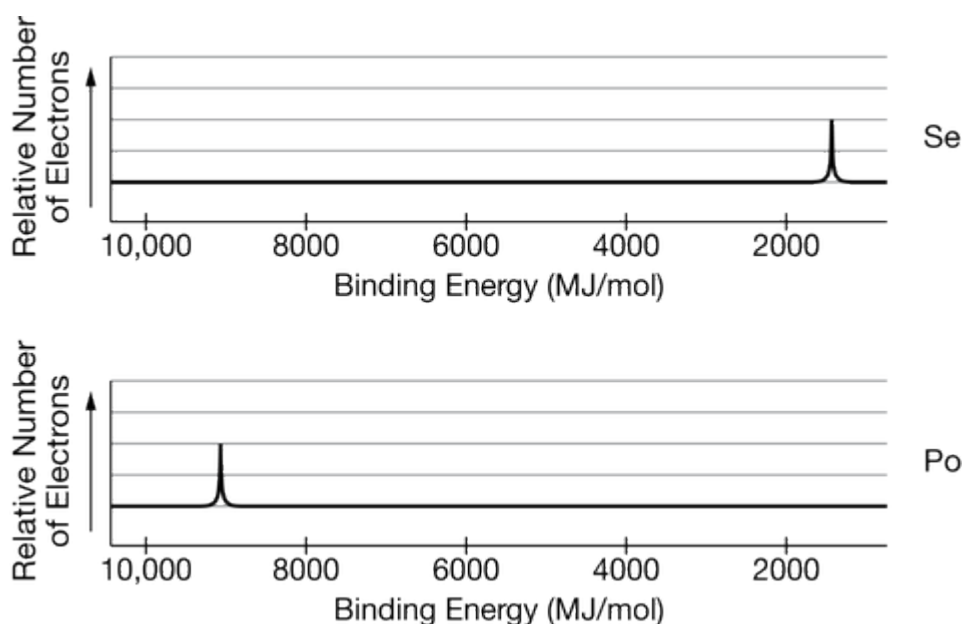


Chapter 04 Part 1: Changes and Representations

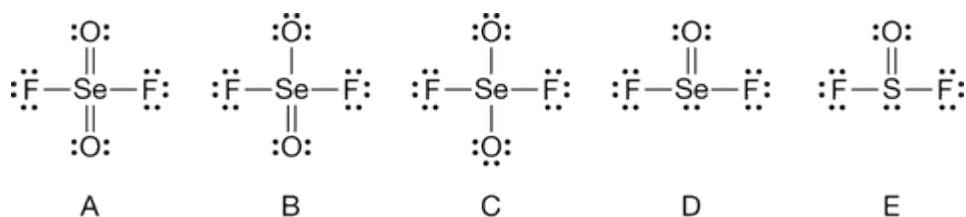
27. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 16 of the periodic table.

- (a) Using principles of atomic structure, explain why the first ionization energy of Po is less than that of Se.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Se and the $1s$ sublevel of Po are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Po is positioned so far to the left of the peak for Se.



Three compounds that contain elements from Group 16 are SeO_2F_2 , SeOF_2 , and SOF_2 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, A, B, or C, minimizes the formal charges for each atom in SeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Se} - \text{to} - \text{O}$ bond in the structure in diagram D is greater than that of the $\text{Se} - \text{to} - \text{O}$ bond in the structure in diagram C. Do you agree or disagree with the student's claim? Justify your answer.

Chapter 04 Part 1: Changes and Representations

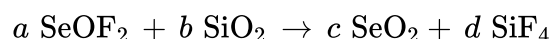
(e) The dipole moment of the SeO_2F_2 molecule (diagrams A, B and C) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of SeOF_2 (diagram D) is greater than the melting point of SOF_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) SeOF_2 reacts with SiO_2 to produce SeO_2 and SiF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The average distance between valence electrons and the nucleus is greater in Po than in Se because Po has more occupied shells. The valence electrons experience a weaker attraction to the nucleus in Po than in Se; therefore, it requires less energy to remove an electron from Po than from Se.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The attraction between $1s$ electrons and the nucleus is stronger in Po than in Se because the Po nucleus has 84 protons and the Se nucleus has 34 protons. According to Coulomb's law, the attractive force is stronger when the magnitude of

Chapter 04 Part 1: Changes and Representations

the charges of the particles is greater. Consequently, the 1s electrons in Po are more strongly attracted to the nucleus (more tightly bound) than those in Se.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

- Structure A minimizes formal charge because it is the only structure in which the formal charge of every atom is zero.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

- Agree. The Se — O bond in structure D is a double bond, whereas the Se — O bond in structure C is a single bond. (It requires more energy to break a Se = O than a Se — O single bond.)

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct geometry and a valid explanation equivalent to the following.

- The molecular geometry is tetrahedral. The Se — O and Se — F bond dipoles are different in magnitude and cannot cancel given this geometric arrangement.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response meets both of the following criteria.

- ☐ The response states that both substances have London dispersion forces
- ☐ The response states that both substances have dipole-dipole attractions

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to one of the following.

· SeOF_2 has a larger, more polarizable electron cloud than SOF_2 due to the greater number of electrons in Se, leading to stronger London dispersion forces. Thus, the attractions between SeOF_2 molecules are stronger than those between SOF_2 molecules, resulting in a higher melting point for SeOF_2 .

OR

· The bonds in SeOF_2 are more polar than those in SOF_2 because the electronegativity difference between Se and oxygen or fluorine is greater than that between S and oxygen or fluorine. Thus, the dipole-dipole attractions between SeOF_2 molecules are stronger than those between SOF_2 molecules, resulting in a higher melting point for SeOF_2 .

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets all of the following criteria.

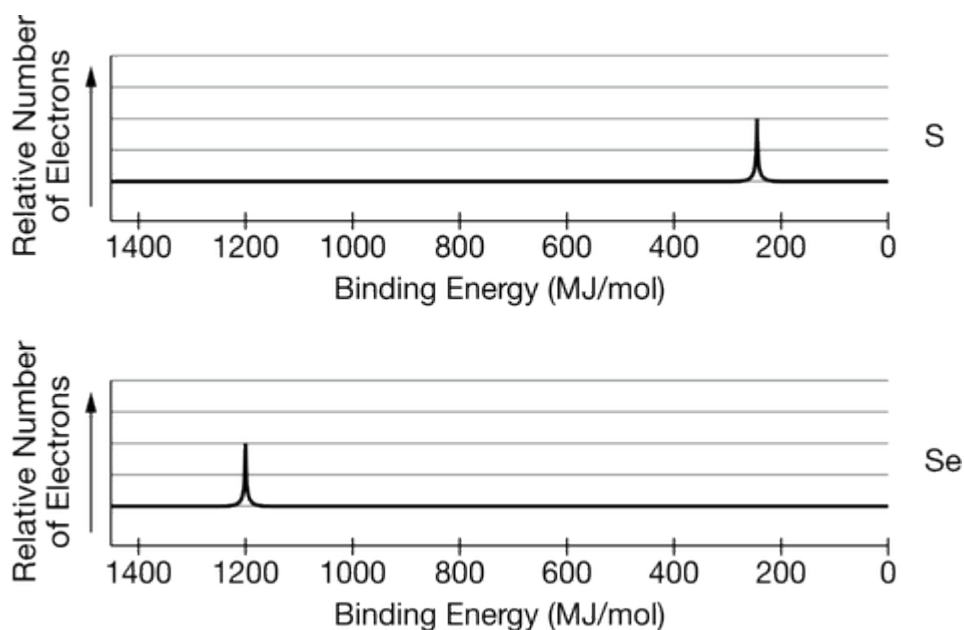
- ☐ The response states that $a = 2$
- ☐ The response states that $b = 1$
- ☐ The response states that $c = 2$
- ☐ The response states that $d = 1$

Chapter 04 Part 1: Changes and Representations

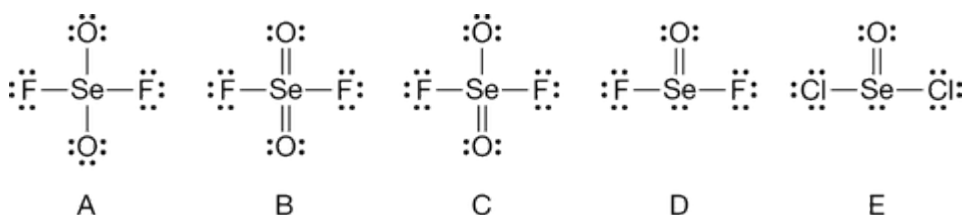
28. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 16 of the periodic table.

- (a) Using principles of atomic structure, explain why the atomic radius of Kr is less than the ionic radius of Se^{2-} .
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of S and the $1s$ sublevel of Se are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Se is positioned so far to the left of the peak for S.



Three compounds that contain elements from Group 16 are SeO_2F_2 , SeOF_2 , and SeOCl_2 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, A, B, or C, minimizes the formal charges for each atom in SeO_2F_2 ? Justify your choice.

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(d) A student proposes that the energy of the Se — to — O bond in the structure in diagram D is less than that of the Se — to — O bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify your answer.

(e) The dipole moment of the SeOF₂ molecule (diagram D) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of SeOF₂ (diagram D) is less than the boiling point of SeOCl₂ (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) SeOF₂ reacts with NaOH to produce Na₂SeO₃, NaF, and H₂O. The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

$e =$

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response provides a valid explanation equivalent to the following.

·Kr and Se^{2−} have the same number of electrons, but the nucleus of Kr has two more protons. Thus, the valence electrons in Kr experience greater attraction to the nucleus compared to Se^{2−} and are, on average, closer to the nucleus.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The attraction between $1s$ electrons and the nucleus is stronger in Se than in S because the Se nucleus has 34 protons and the S nucleus has 16 protons. According to Coulomb's law, the attractive force is stronger when the magnitude of the charges of the particles is greater. Consequently, the $1s$ electrons in Se are more strongly attracted to the nucleus (more tightly bound) than those in S.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct choice and a valid justification equivalent to the following.

· Structure B minimizes formal charge because it is the only structure in which the formal charge of every atom is zero.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

· Disagree. The Se — O bond in structure D is a double bond, which has a greater energy compared to the Se — O bond in structure A, which is a single bond. (It requires more energy to break a $\text{Se} = \text{O}$ than a $\text{Se} - \text{O}$ single bond.)

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct geometry and a valid explanation equivalent to the following.

· The molecular geometry is trigonal pyramidal. Given the asymmetric arrangement of atoms relative to the lone pair of electrons, it is impossible for all of the individual bond dipoles to cancel completely.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response meets both of the following criteria.

- ☐ The response states that both substances have London dispersion forces
- ☐ The response states that both substances have dipole-dipole attractions

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· SeOF_2 has fewer electrons than SeOCl_2 , thus a smaller, less polarizable electron cloud. Thus, the attractions between SeOF_2 molecules are weaker than those between SeOCl_2 molecules, resulting in a lower boiling point.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets all of the following criteria.

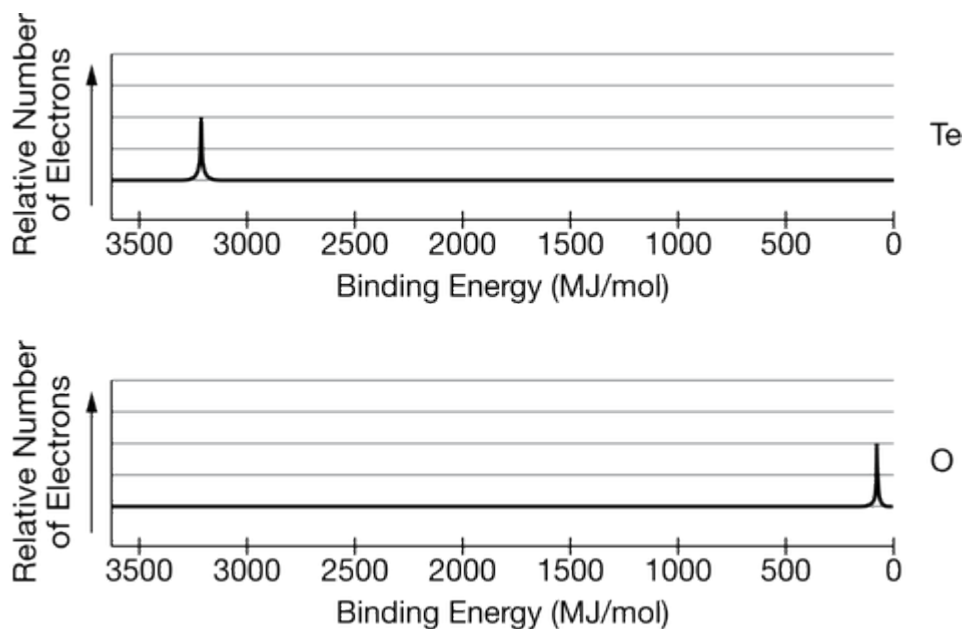
- ☐ The response states that $a = 1$
- ☐ The response states that $b = 4$
- ☐ The response states that $c = 1$
- ☐ The response states that $d = 2$
- ☐ The response states that $e = 2$

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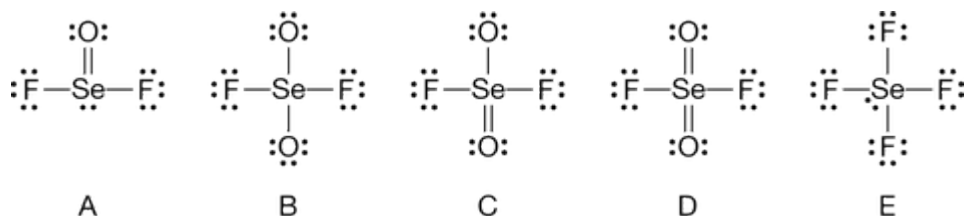
29. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 16 of the periodic table.

- (a) Using principles of atomic structure, explain why the atomic radius of Te is less than the ionic radius of Te^{2-} .
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Te and the $1s$ sublevel of O are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for O is positioned so far to the right of the peak for Te.



Three compounds that contain elements from Group 16 are SeOF_2 , SeO_2F_2 , and SeF_4 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, B, C or D, minimizes the formal charges for each atom in SeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Se} - \text{to} - \text{F}$ bond in the structure in diagram A is equal to that of

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the Se — to — F bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

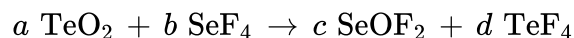
(e) The dipole moment of the SeF₄ molecule (diagram E) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of SeOF₂ (diagram A) is greater than the melting point of SeO₂F₂ (diagrams B, C, and D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) TeO₂ reacts with SeF₄ to produce SeOF₂ and TeF₄. The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· Te and Te^{2−} have the same number of protons (same nuclear charge), but Te has two fewer electrons. Therefore, the valence electrons in Te are, on average, closer to the nucleus because they experience fewer electron-electron repulsions within the same shell.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

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· The attraction between 1 *s* electrons and the nucleus is weaker in O than in Te because the O nucleus has 8 protons while the Te nucleus has 52 protons. According to Coulomb's law, the attractive force is weaker when the magnitude of the charges of the particles is smaller. Consequently, the 1 *s* electrons in O are less strongly attracted to the nucleus (less tightly bound) than those in Te.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct choice and a valid justification equivalent to the following.

· Structure D minimizes formal charge because it is the only structure in which the formal charge of every atom is zero.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

· Agree. The Se — F bond in both structures is a single bond. (Similar single bonds should require the same amount of energy to break.)

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct geometry and a valid explanation equivalent to the following.

· The molecular geometry is see-saw. Given the asymmetric arrangement of atoms relative to the lone pair of electrons, the individual bond dipoles cannot cancel completely.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response meets both of the following criteria.

- ☐ The response states that both substances have London dispersion forces
- ☐ The response states that both substances have dipole-dipole attractions

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· SeOF_2 has a trigonal pyramidal geometry, resulting in an asymmetrical arrangement of three bond dipoles that do not cancel. This creates a greater dipole moment than in SeO_2F_2 , where the partially opposing dipoles of the two $\text{Se} - \text{O}$ bonds and the two $\text{Se} - \text{F}$ bonds result in a smaller dipole moment. Thus, the dipole-dipole attractions between SeOF_2 molecules are stronger than those between SeO_2F_2 molecules, resulting in a higher melting point for SeOF_2 .

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets all of the following criteria.

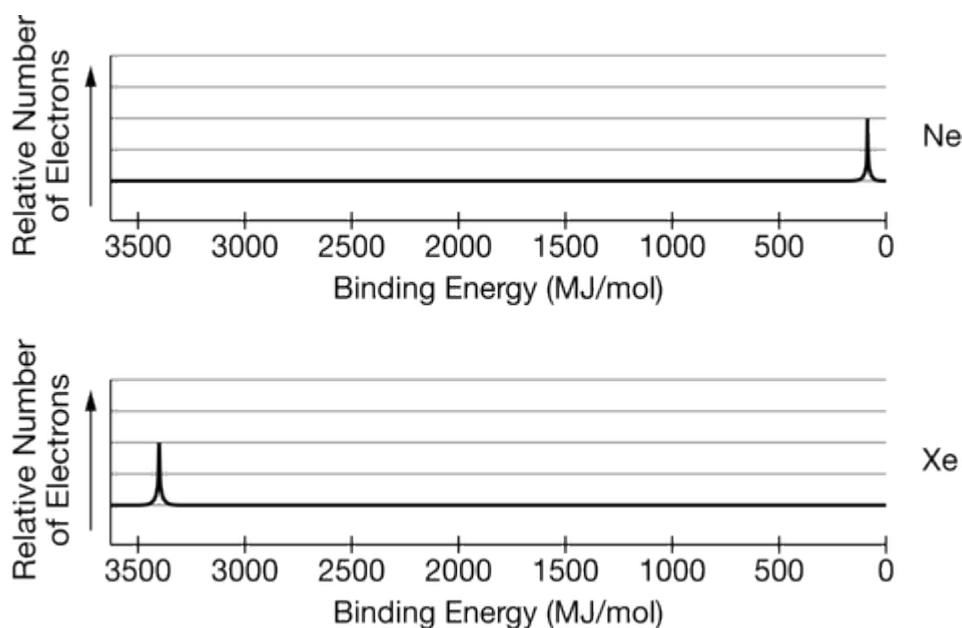
- ☐ The response states that $a = 1$
- ☐ The response states that $b = 2$
- ☐ The response states that $c = 2$
- ☐ The response states that $d = 1$

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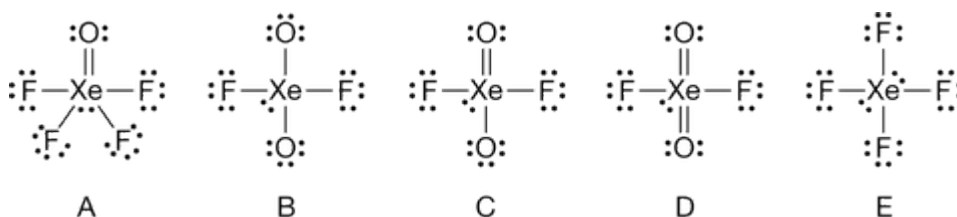
30. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 18 of the periodic table.

- (a) Using principles of atomic structure, explain why the atomic radius of Xe is less than that of Te.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Ne and the $1s$ sublevel of Xe are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Ne is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are XeOF_4 , XeO_2F_2 , and XeF_4 . Possible Lewis electron dot diagrams for the compounds are shown below.



- (c) Which structure, B, C, or D, minimizes the formal charges for each atom in XeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the $\text{Xe}-\text{to}-\text{F}$ bond in the structure in diagram B is equal to that of the $\text{Xe}-\text{to}-\text{F}$ bond in the structure in diagram A. Do you agree or disagree with the student's claim? Justify

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your answer.

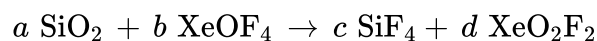
(e) The dipole moment of the XeOF_4 molecule (diagram A) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The boiling point of XeF_4 (diagram E) is greater than the boiling point of elemental Xe.

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in boiling points based on the types and relative strengths of intermolecular forces.

(h) XeOF_4 reacts with SiO_2 to produce XeO_2F_2 and SiF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The valence electrons in Te and Xe are in the $n = 5$ shell; however, Xe has a greater number of protons in its nucleus than Te. Thus, the electrons in Xe experience a greater attraction to the nucleus compared to Te and are, on average, closer to the nucleus.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The attraction between 1 s electrons and the nucleus is weaker in Ne than in Xe because the Ne nucleus has 10 protons

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while the Xe nucleus has 54 protons. According to Coulomb's law, the attractive force is weaker when the magnitude of the charges of the particles is smaller. Consequently, the 1 *s* electrons in Ne are less strongly attracted to the nucleus (less tightly bound) than those in Xe.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct choice and a valid justification equivalent to the following.

- Structure D minimizes formal charge because it is the only structure in which the formal charge of every atom is zero.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

- Agree. The Xe — F bond in structure B and in structure A are both single bonds. (Single bonds between the same atoms should require the same amount of energy to break.)

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct geometry and a valid explanation equivalent to the following.

- The molecular geometry is square pyramidal. Given the asymmetric arrangement of atoms relative to the lone pair of electrons, it is impossible for all of the individual bond dipoles to cancel completely.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response states that both substances have London dispersion forces.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

XeF_4 is a nonpolar substance and Xe is an atomic element. Both substances experience only London dispersion forces, but XeF_4 has more electrons and therefore a larger, more polarizable electron cloud than Xe, resulting in stronger total intermolecular forces than Xe. As a result, XeF_4 has the higher boiling point.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets all of the following criteria.

- ☐ The response states that $a = 1$
- ☐ The response states that $b = 2$
- ☐ The response states that $c = 1$
- ☐ The response states that $d = 2$

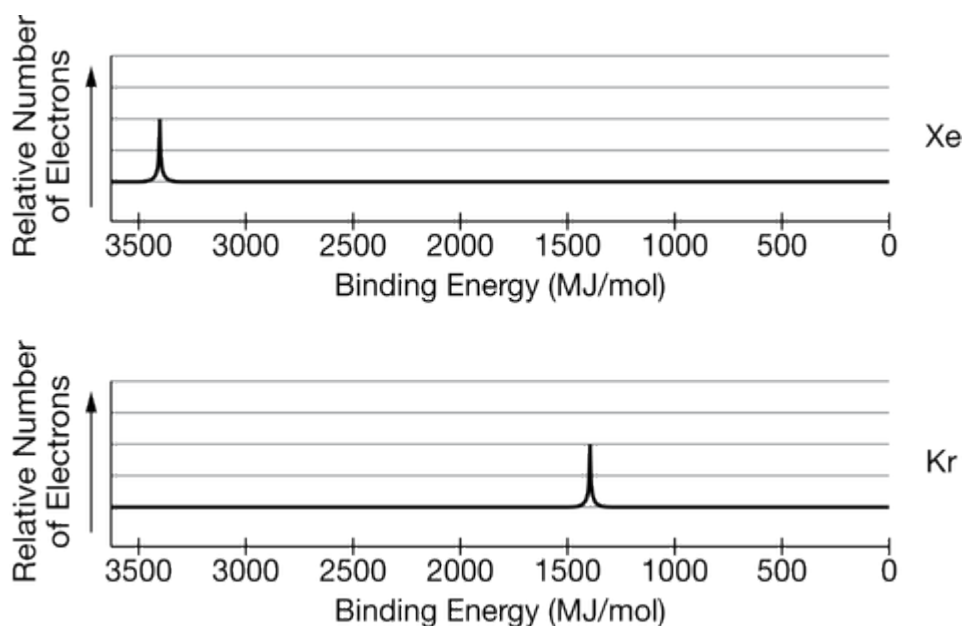
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31. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

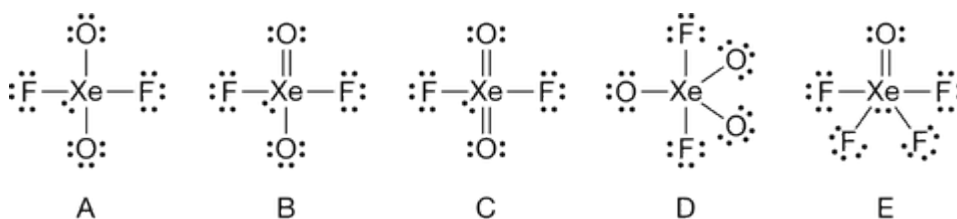
Answer the following questions about elements in Group 18 of the periodic table.

(a) Using principles of atomic structure, explain why the ionic radius of Cs^+ is less than the atomic radius of Xe.

(b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Kr are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Kr is positioned so far to the right of the peak for Xe.



Three compounds that contain an element from Group 18 are XeO_2F_2 , XeO_3F_2 , and XeOF_4 . Possible Lewis electron dot diagrams for the compounds are shown below.



(c) Which structure, A, B, or C, minimizes the formal charges for each atom in XeO_2F_2 ? Justify your choice.

(d) A student proposes that the energy of the $\text{Xe} - \text{O} - \text{O}$ bond in the structure in diagram E is greater than that of the $\text{Xe} - \text{O} - \text{O}$ bond in the structure in diagram D. Do you agree or disagree with the student's claim? Justify your answer.

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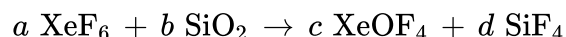
(e) The dipole moment of the XeO_3F_2 molecule (diagram D) is zero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a zero dipole moment.

The melting point of XeOF_4 (diagram E) is greater than the melting point of XeO_3F_2 (diagram D).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeF_6 reacts with SiO_2 to produce XeOF_4 and SiF_4 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· Cs^+ and Xe have the same number of electrons, but Cs^+ has one more proton than Xe. Thus, the valence electrons in Cs^+ experience greater attraction to the nucleus compared to Xe and are, on average, closer to the nucleus.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The attraction between 1 s electrons and the nucleus is weaker in Kr than in Xe because the Kr nucleus has 36 protons while the Xe nucleus has 54 protons. According to Coulomb's law, the attractive force is weaker when the magnitude of the charges of the particles is smaller. Consequently, the 1 s electrons in Kr are less strongly attracted to the nucleus (less tightly bound) than those in Xe.

Chapter 04 Part 1: Changes and Representations**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct choice and a valid justification equivalent to the following.

- Structure C minimizes formal charge because it is the only structure in which the formal charge of every atom is zero.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

- Agree. The Xe — O bond in structure E is a double bond compared to the Xe — O bond in structure D, which is a single bond. (It requires more energy to break a Xe = O than a Xe — O single bond.)

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct geometry and a valid explanation equivalent to the following.

- The molecular geometry is trigonal bipyramidal. Given this arrangement of atoms, the individual Xe — F bond dipoles cancel and the individual Xe — O bond dipoles cancel, resulting in a zero dipole moment.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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Chapter 04 Part 1: Changes and Representations

The response states that both substances have London dispersion forces.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

Both substances experience comparable London dispersion forces, but XeOF_4 additionally has dipole-dipole attractions. The total intermolecular forces in XeOF_4 are stronger than those in XeO_3F_2 , leading to a higher melting point for XeOF_4 .

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets all of the following criteria.

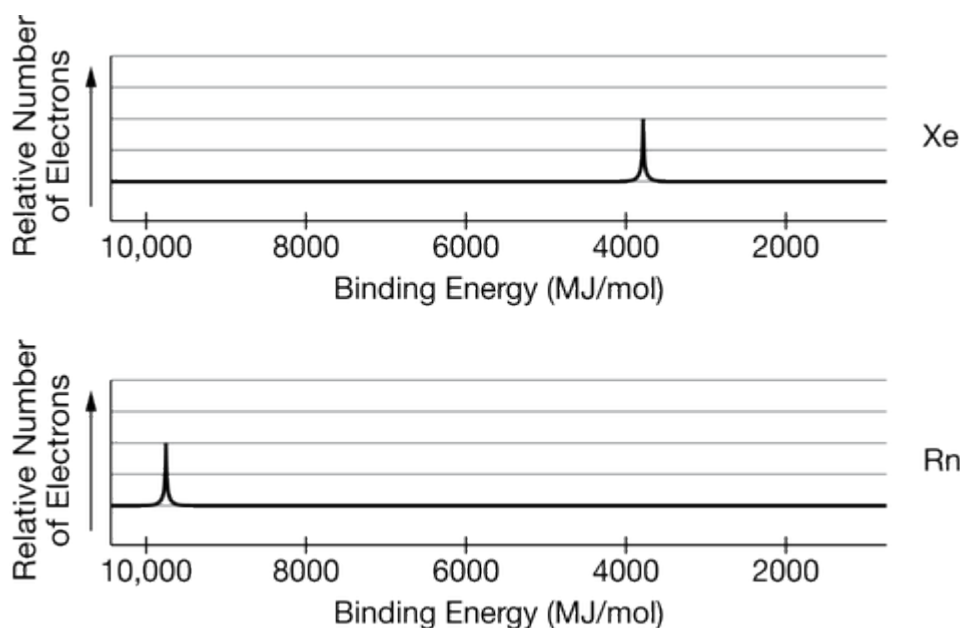
- ☐ The response states that $a = 2$
- ☐ The response states that $b = 1$
- ☐ The response states that $c = 2$
- ☐ The response states that $d = 1$

Chapter 04 Part 1: Changes and Representations

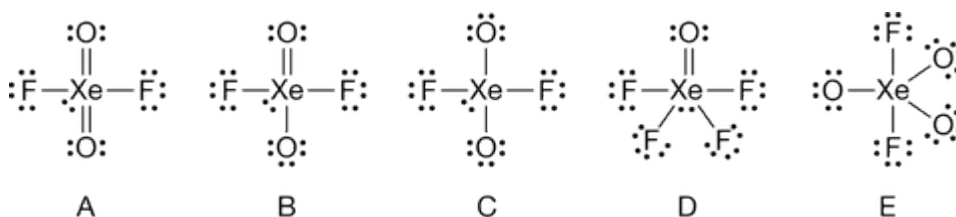
32. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about elements in Group 18 of the periodic table.

- (a) Using principles of atomic structure, explain why the first ionization energy of Rn is less than that of Xe.
- (b) Photoelectron spectroscopy data for the $1s$ sublevel of Xe and the $1s$ sublevel of Rn are represented below. In terms of Coulomb's law and atomic structure, explain why the peak for Xe is positioned so far to the right of the peak for Rn.



Three compounds that contain an element from Group 18 are XeO_2F_2 , XeOF_4 , and XeO_3F_2 . Possible Lewis electron-dot diagrams for the compounds are shown below.



- (c) Which structure, A, B or C, minimizes the formal charges for each atom in XeO_2F_2 ? Justify your choice.
- (d) A student proposes that the energy of the Xe — to — O bond in the structure in diagram D is less than that of the Xe — to — O bond in the structure in diagram E. Do you agree or disagree with the student's claim? Justify

Chapter 04 Part 1: Changes and Representations

your answer.

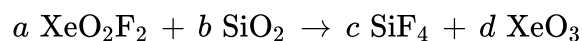
(e) The dipole moment of the XeO_2F_2 molecule (diagrams A, B and C) is nonzero. Identify its molecular geometry and explain how its geometry and the polarity of its bonds result in a nonzero dipole moment.

The melting point of XeO_2F_2 (diagrams A, B, C) is greater than the melting point of XeO_3F_2 (diagram E).

(f) Identify the type(s) of intermolecular force(s) that the two substances have in common.

(g) Explain the difference in melting points based on the types and relative strengths of intermolecular forces.

(h) XeO_2F_2 reacts with SiO_2 to produce SiF_4 and XeO_3 . The unbalanced chemical equation for the reaction appears below.



Give the value of each of the following coefficients in the balanced equation.

$a =$

$b =$

$c =$

$d =$

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The valence electrons in Rn on average are further from the nucleus than in Xe because Rn has more occupied shells. The valence electrons experience a weaker attraction to the nucleus in Rn than in Xe; therefore, it requires less energy to remove a valence electron from Rn than from Xe.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

· The attraction between 1 s electrons and the nucleus is weaker in Xe than in Rn because the Xe nucleus has 54 protons

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while the Rn nucleus has 86 protons. According to Coulomb's law, the attractive force is weaker when the magnitude of the charges of the particles is smaller. Consequently, the 1 *s* electrons in Xe are less strongly attracted to the nucleus (less tightly bound) than those in Rn.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct choice and a valid justification equivalent to the following.

- Structure A minimizes formal charge because it is the only structure in which the formal charge of every atom is zero.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

- Disagree. The Xe — O bond in structure D is a double bond, which has a greater energy compared to the Xe — O bond in structure E, which is a single bond. (It requires more energy to break a Xe = O than a Xe — O single bond.)

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct geometry and a valid explanation equivalent to the following.

- The molecular geometry is see-saw. Given the asymmetric arrangement of atoms relative to the lone pair of electrons, it is impossible for all of the individual bond dipoles to cancel completely.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response states that both substances have London dispersion forces.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

Both substances experience comparable London dispersion forces, but XeO_2F_2 additionally has dipole-dipole attractions. The total intermolecular forces among molecules of XeO_2F_2 are stronger than those among molecules of XeO_3F_2 , leading to a higher melting point for XeO_2F_2 .

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets all of the following criteria.

- ☐ The response states that $a = 2$
- ☐ The response states that $b = 1$
- ☐ The response states that $c = 1$
- ☐ The response states that $d = 2$